CARDIOVASCULAR RISK PREDICTION USING AN OPTIMIZED XGBOOST AND DEEP LEARNING-BASED ECG CLASSIFICATION SYSTEM

A PROJECT REPORT

Submitted by

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in partial fulfillment of the requirements for the award degree of

Bachelor in Engineering

20CS8504 - PROJECT WORK

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

K. RAMAKRISHNAN COLLEGE OF TECHNOLOGY

(AUTONOMOUS)

SAMAYAPURAM – 621112

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BONAFIDE CERTIFICATE

The work embodied in the present project report entitled "**CARDIOVASCULAR RISK PREDICTION USING AN OPTIMIZED XGBOOST AND DEEP LEARNING-BASED ECG CLASSIFICATION SYSTEM**" has been carried out by the students **SERALATHAN S, UMA MAHESWARI M, VIJAY V, VINOTH T**, The work reported herein is original and we declare that the project is their own work, except where specifically acknowledged, and has not been copied from other sources or been previously submitted for assessment.

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ABSTRACT

Cardiovascular Risk Prediction and ECG Classification System is an intelligent deep-learning–based system designed to support early identification and awareness of cardiovascular diseases, including coronary artery disease, arrhythmias, myocardial infarction, and other clinically significant cardiac conditions. Early detection of such conditions is critical, as delayed diagnosis often leads to increased morbidity, higher treatment costs, and preventable mortality. However, conventional cardiovascular risk assessment relies heavily on manual clinical evaluation and specialist expertise for ECG interpretation, which is prone to subjectivity, time-consuming, and often inaccessible in rural or resource-constrained environments. The system addresses these challenges through two integrated AI modules: an Optimized XGBoost model that analyzes structured patient clinical parameters to predict cardiovascular risk, and a hybrid CNN-LSTM deep learning model that classifies ECG signals to detect various heart rhythms and arrhythmias.

Keywords: Artificial Intelligence (Ai), Cardiovascular Risk Prediction, Ecg Classification, Xgboost, Hybrid Cnn-Lstm Deep Learning, Explainable Ai (Shap), Arrhythmia Detection, Clinical Decision Support System**.**

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SIGNATURE

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LIST OF SYMBOLS AND ABBREVIATIONS

ANN	-	Artificial Newrl Network
API	-	Application Programming Interface
CNN	-	Convolutional Neural Network
DDPM	-	Denoising Diffusion Probabilistic Model
GRAD-CAM	-	Gradient-Weighted Class Activation Mapping
HDNN	-	Hybrid Deep Neural Network
KNN	-	K-Nearest Neighbors
LIME	-	Local Interpretable Model-Agnostic Explanations
LLM	-	Large Language Model
LSTM	-	Long Short-Term Memory
MRMR	-	Minimal Redundancy Maximal Relevance
PCA	-	Principal Component Analysis
ReLU	-	Rectified Linear Unit
SVM	-	Support Vector Machine
XAI	-	Explainable Artificial Intelligence
XGBoost	-	Extreme Gradient Boosting

CHAPTER 1

INTRODUCTION

1.1. DESCRIPTION

Heart disease is one of the most critical and complex public health challenges of the twenty-first century. Cardiovascular diseases encompass a broad spectrum of conditions affecting the heart and blood vessels, including coronary artery disease, arrhythmias, heart failure, valvular heart disease, and congenital heart defects. Despite significant advances in modern medicine, cardiovascular disease remains the number one cause of death globally, responsible for approximately 32% of all global deaths annually. The WHO estimates that more than 17.9 million lives are lost to cardiovascular events each year, with low- and middle-income countries bearing a disproportionately high burden of this disease.

The primary risk factors for cardiovascular disease are well established and include hypertension, hyperlipidemia, diabetes mellitus, obesity, physical inactivity, tobacco use, and a family history of heart disease. However, the interplay between these multiple risk factors is highly complex and non-linear, making accurate individual risk prediction challenging through traditional clinical scoring systems such as the Framingham Risk Score. These conventional approaches are limited in their ability to capture the complex relationships between risk factors and fail to leverage the predictive potential available in large-scale electronic health records and multimodal physiological data.

The Electrocardiogram (ECG) is a fundamental, non-invasive, and cost-effective diagnostic tool that records the electrical activity of the heart over time. ECG analysis provides critical clinical information for diagnosing a wide range

of cardiac conditions, including myocardial infarction, atrial fibrillation, ventricular tachycardia, bundle branch blocks, and ST-segment abnormalities. However, the accurate interpretation of ECG signals requires extensive clinical training and expertise, and even experienced cardiologists exhibit significant inter-observer variability in ECG reading, particularly for complex or subtle rhythm abnormalities. This variability contributes to misdiagnosis and delayed treatment, with potentially serious clinical consequences.

The rapid advancement of Artificial Intelligence, Machine Learning, and Deep Learning technologies has created transformative opportunities for automating and enhancing cardiovascular risk prediction and ECG interpretation. Machine learning algorithms can identify complex, non-linear patterns in high-dimensional clinical datasets that are beyond the capacity of human analysis. Deep learning architectures, particularly Convolutional Neural Networks and Recurrent Neural Networks, have demonstrated remarkable performance in automatic ECG signal classification, achieving accuracy levels comparable to or exceeding those of board-certified cardiologists on standard benchmark datasets.

1.2. OVERVIEW

The Cardiovascular Risk Prediction Using an Optimized XGBoost and Deep Learning-Based ECG Classification System is an integrated, intelligent clinical decision support platform that addresses the limitations of traditional cardiovascular diagnosis methods. The system combines two complementary AI-driven analysis pipelines: a structured data-based cardiovascular risk prediction module powered by an optimized XGBoost algorithm, and a signal-based ECG classification module powered by a hybrid CNN-LSTM deep learning architecture.

The structured data prediction module accepts a comprehensive set of patient clinical parameters as input and generates a cardiovascular risk assessment indicating the probability of heart disease. The system employs Bayesian optimization for XGBoost hyperparameter tuning, ensuring that the

model achieves its maximum predictive performance. SHAP-based Explainable AI techniques are integrated to generate feature importance explanations, making the prediction results transparent and interpretable for clinical users.

The ECG classification module processes uploaded ECG signal data through a multi-stage deep learning pipeline. A one-dimensional CNN extracts local morphological features from the ECG waveform, while a bidirectional LSTM network captures temporal dependencies and sequential patterns in the signal. An attention mechanism focuses the model on the most diagnostically relevant segments of the ECG. The classification output identifies the cardiac rhythm type and flags any detected arrhythmias or ST-segment abnormalities. SHAP and Gradient-weighted Class Activation Mapping (Grad-CAM) techniques provide interpretable visualizations of the ECG segments that contributed most significantly to the classification decision.

1.3. PROBLEM STATEMENT

Cardiovascular disease prediction and ECG-based cardiac rhythm classification currently face several significant challenges that limit the effectiveness and accessibility of existing diagnostic systems. Manual forensic analysis of patient health parameters by clinicians is time-consuming and subject to cognitive biases and fatigue-related errors. Clinical risk scoring tools such as the Framingham Risk Score and SCORE2 provide only binary or categorical risk stratification based on a limited set of pre-selected linear predictors, failing to capture the complex, non-linear interactions between multiple cardiovascular risk factors present in real-world patient populations.

ECG interpretation remains a highly specialized skill that is unevenly distributed across healthcare settings. Primary care physicians and general practitioners often lack the specialist training required to accurately diagnose complex arrhythmias or subtle ST-segment abnormalities from ECG tracings. This expertise gap is particularly acute in rural areas, low-resource healthcare

facilities, and emergency settings where immediate ECG interpretation is clinically critical but cardiologist consultation is unavailable.

As a result, delays in accurate diagnosis can lead to inappropriate or delayed treatment, increasing the risk of adverse patient outcomes. Inconsistent interpretation may also contribute to variability in clinical decisions and reduced confidence among healthcare providers. Therefore, there is a growing need for automated and intelligent ECG analysis systems to support timely, accurate, and reliable diagnosis across diverse healthcare environments.

Existing automated ECG analysis systems are frequently black-box models that provide classification outputs without explaining the clinical reasoning behind their decisions. This lack of transparency creates significant barriers to clinical adoption, as healthcare professionals are understandably reluctant to act on unexplained AI recommendations when making high-stakes clinical decisions affecting patient safety. There is therefore a critical need for an accurate, transparent, and interpretable AI-based system that can support cardiovascular risk prediction and ECG classification in diverse clinical environments.

1.4. OBJECTIVES

The primary aim of this project is to develop an integrated, AI-powered cardiovascular risk prediction and ECG classification system that delivers accurate, transparent, and clinically actionable assessments to support healthcare decision-making. The specific objectives are:

1. To develop an optimized XGBoost-based machine learning model for cardiovascular risk prediction using structured patient clinical parameters.
2. To implement Bayesian hyperparameter optimization to maximize the predictive performance of the XGBoost model on the UCI Heart Disease Dataset.

3. To design and train a hybrid CNN-LSTM deep learning architecture for multi-class ECG signal classification, targeting the detection of normal sinus rhythm and common cardiac arrhythmias.
4. To integrate SHAP-based Explainable AI techniques into both modules to provide transparent, interpretable explanations of prediction and classification outputs.
5. To develop a secure, web-based application using Python and Flask that provides an intuitive interface for cardiovascular risk assessment and ECG classification.
6. To implement a comprehensive visualization module using Matplotlib and Seaborn for graphical presentation of prediction results, SHAP feature importance plots, and ECG classification confidence scores.
7. To provide a downloadable clinical report generation feature for documentation and referral purposes.
8. To evaluate the performance of the proposed system against existing state-of-the-art cardiovascular risk prediction and ECG classification methods using standard benchmark datasets.

1.5. USERS OF THE SYSTEM

The proposed system is designed to serve the following categories of users across the healthcare ecosystem:

1. Cardiologists and Cardiac Specialists: Use the ECG classification module to support automated pre-screening of large volumes of ECG recordings, flagging anomalies for specialist review and reducing diagnostic workload.
2. General Practitioners and Primary Care Physicians: Use the cardiovascular risk prediction module to assess individual patient risk profiles based on routine clinical parameters and decide on appropriate referrals or preventive interventions.

3. Healthcare Researchers: Use the system to analyze population-level cardiovascular risk patterns and evaluate the clinical utility of different risk factors in diverse patient cohorts.
4. Hospital Administrators and Health IT Personnel: Manage system deployment, user accounts, patient record databases, and integration with existing hospital information systems.
5. Patients and Health-Conscious Individuals: In a patient-facing deployment context, use the system to assess their individual cardiovascular risk based on available health parameters and receive guidance on risk reduction.

1.6. APPLICATION FIELDS

The proposed cardiovascular risk prediction and ECG classification system has broad applicability across multiple domains:

1. Clinical Decision Support in Hospitals and Clinics: Real-time cardiovascular risk assessment and ECG interpretation assistance for attending physicians.
2. Telemedicine and Remote Patient Monitoring: Remote cardiovascular screening for patients in geographically isolated areas or those with limited mobility.
3. Emergency Medicine: Rapid ECG classification for triage and priority-setting in emergency departments.
4. Preventive Healthcare Programs: Population-wide cardiovascular risk screening in community health programs and employer wellness initiatives.
5. Medical Education and Training: As a teaching tool for medical students and residents learning ECG interpretation and cardiovascular risk assessment.
6. Insurance and Actuarial Applications: Cardiovascular risk stratification for health insurance underwriting and premium determination.

1.7. SCOPE OF THE PROJECT

The scope of this project encompasses the design, development, testing, and validation of an integrated cardiovascular risk prediction and ECG classification system using advanced machine learning and deep learning techniques. The system supports binary cardiovascular risk classification (high risk versus low risk) based on structured clinical parameters, and multi-class ECG rhythm classification covering normal sinus rhythm, atrial fibrillation, ventricular tachycardia, ST-elevation myocardial infarction, and bundle branch block patterns. The system is validated on publicly available benchmark datasets including the UCI Heart Disease Dataset and the MIT-BIH Arrhythmia Database. Future extensions may include integration with wearable ECG devices, real-time streaming analysis, and federated learning for privacy-preserving model training across distributed hospital networks.

CHAPTER 2

LITERATURE SURVEY

2.1. ADVANCED HEART ATTACK RISK PREDICTION USING STACKED HYBRID MACHINE LEARNING IN 2025

Rudraksh Singh Bhaduaria et al**.** in this paper tells that, heart disease has been one of the leading causes of death worldwide, and early detection forms the basis of effective intervention and treatment. The study presents an all-inclusive machine learning framework developed for predicting the risk of heart disease by using many advanced classification techniques. The authors argue that no single classifier can reliably generalize across diverse patient populations, motivating the development of hybrid and stacked ensemble approaches. The proposed framework implements and compares hybrid models that combine algorithms like Random Forest (RF), Support Vector Machine (SVM), XGBoost, and Logistic Regression (LR) for predictive performance improvement. These classifiers are evaluated both individually and in stacked ensemble configurations to identify the most effective combination for heart attack risk prediction. Traditional models like K-Nearest Neighbors (KNN), Naive Bayes, and Decision Trees are also included as baseline comparisons to contextualize the improvement offered by hybrid approaches. A key methodological contribution of this work is the thorough application of feature importance analysis using Random Forest, which is identified as a critical step towards identifying the most relevant predictors in the presence of heart disease. The authors demonstrate that removing low-importance features and focusing on a reduced but highly informative feature set leads to both improved accuracy and faster inference times, making the system more practical for clinical deployment.

2.2. CARDIAC CLARITY: HARNESSING MACHINE LEARNING FOR ACCURATE HEART DISEASE PREDICTION IN 2025

Sujith Santhosh et al. in this paper tells that, the major contributor to global mortality is cardiovascular disease, posing a formidable challenge to the global healthcare system. Heart disease often develops and progresses without noticeable symptoms, emphasizing the need for earlier detection to prevent severe outcomes such as myocardial infarction and cardiac arrest. The study acknowledges that AI models provide powerful tools for accurate diagnosis, data analysis, and predictive assessment of individuals who may be at risk of heart disease. A particularly important aspect of this work is the emphasis on Explainable AI (XAI), which demystifies the decisions of AI models, making them transparent and understandable to healthcare professionals. The authors argue that explainability is not an optional feature but a clinical necessity, as it enables doctors to validate AI outputs, identify potential biases, and maintain responsibility for final treatment decisions. XAI techniques such as SHAP, LIME, QLattice, and Anchor are studied and applied to provide post-hoc explanations of the model predictions. The dataset analyzed in this study was collected from a multispecialty hospital in India and is publicly available on the Mendeley data repository. This real-world clinical dataset contains a wide range of attributes including patient demographics, laboratory measurements, lifestyle factors, and cardiac examination results. Hyperparameter tuning was performed using three distinct strategies: Grid Search, Random Search, and Bayesian Optimization. This multi-strategy tuning approach ensures that the final model parameters are globally optimized rather than locally trapped. The study concludes by providing a reliable and transparent framework for early detection of heart disease that enables patients to receive timely and personalized care. The integration of XAI into the prediction pipeline is particularly relevant to the proposed project.

2.3. HEART DISEASE CLASSIFICATION FROM ECG IMAGES USING DEEP LEARNING IN 2025

Muhammad Raoof et al. in this paper tells that, heart disease remains one of the leading causes of death globally, with cardiovascular conditions accounting for nearly one-third of all deaths worldwide each year. Echocardiography is one of the most commonly used non-invasive imaging methods to diagnose cardiovascular conditions, providing real-time visualization of the heart's structure, motion, and functional performance. The technique captures dynamic information about cardiac wall motion, valve function, ejection fraction, and ventricular geometry that is essential for clinical assessment. However, accurately interpreting echocardiogram images requires a high level of specialized cardiovascular knowledge and years of clinical training, making it difficult for general practitioners and non-specialist healthcare providers to effectively utilize this diagnostic modality. In resource-constrained healthcare settings, the shortage of qualified echocardiography interpreters leads to diagnostic delays and missed opportunities for early intervention. This study aimed to address this critical challenge by developing advanced deep learning models capable of automatically classifying heart disease based on echocardiogram data. The proposed deep learning model is inspired by the VGG16 architecture, a well-established convolutional neural network known for its depth and feature extraction capability. The experimental results suggest that the proposed method can be effectively used in the automated classification of cardiovascular diseases from echocardiogram images, potentially improving early detection rates and reducing the dependency on specialist interpretation. This study highlights the growing importance of image-based classification in cardiac diagnostics and directly informs the image classification module of the proposed Heart Disease Prediction System, which similarly employs trained ML models to classify uploaded medical images as normal or abnormal.

2.4. INTERPRETABLE HYBRID MULTICHANNEL DEEP LEARNING MODEL FOR HEART DISEASE CLASSIFICATION USING 12-LEAD ECG SIGNAL IN 2024

Yehualashet Megersa Ayano et al. in this paper tells that, an electrocardiogram (ECG) is a widely used, non-invasive and cost-effective method for diagnosing heart disease that records the electrical activity of the heart over a period of time. Despite its widespread clinical use, physicians often face significant challenges in correctly interpreting ECG recordings due to the complexity of the waveform morphology, inter-patient variability, and the presence of noise artifacts in clinical recordings. These challenges motivate the development of automated deep learning systems that can assist clinicians in ECG interpretation. As a result, deep learning (DL) models have been proposed to assist with ECG interpretation over the past several years. However, the development of a robust, interpretable model that performs consistently well across diverse ECG datasets from different patient populations, recording conditions, and equipment specifications remains a major research challenge. Hence, this study presents a robust interpretable DL-based system that addresses both performance and interpretability requirements simultaneously. The classification output is analyzed using two post-hoc model interpretability techniques: Gradient-weighted Class Activation Mapping Plus Plus (Grad-CAM++) and SHAP. These techniques are applied to visualize the influential segments of the ECG signal, both at the instance level for individual samples and at the test set level to assess the global contributions of each of the 12 ECG leads to the model's predictions. The model's combination of high performance and transparent output interpretation makes it a highly practicable tool for ECG-based heart disease diagnosis, and the use of SHAP for interpretability directly aligns with the explainability approach adopted in the proposed Heart Disease Prediction System.

2.5. A ROBUST HEART DISEASE PREDICTION SYSTEM USING HYBRID DEEP NEURAL NETWORKS IN 2023

Mana Saleh Al Reshan et al. in this paper tells that, Heart Disease (HD) is recognized as the leading cause of worldwide mortality by the World Health Organization (WHO), resulting in the loss of approximately 17.9 million lives each year. HD prediction is found to be a challenging issue that can provide a computerized estimate of the level of HD so that additional clinical action can be simplified. Early detection and accurate prediction of HD play a critical role in providing timely medical interventions and improving patient outcomes. Thus, HD prediction has attracted massive attention worldwide in healthcare environments. Deep Learning (DL) based systems have played a significant role in various disease prediction and diagnosis tasks with good efficiency. To this end, the main contribution of this paper is to design a robust HD prediction system using Hybrid Deep Neural Networks (HDNNs) that involves combining multiple neural network architectures to extract and learn relevant features from the input data. The HDNN is employed to apply its feature learning capabilities and non-linear transformation technology to capture complex patterns and relationships in HD datasets, leading to enhanced prediction accuracy. For this purpose, three DL models, namely Artificial Neural Networks (ANN), Convolutional Neural Networks (CNN), Long Short-Term Memory (LSTM), and a new HDNN model combining both CNN and LSTM along with additional Dense layers are proposed to develop the hybrid HD prediction architecture. The HDNN model leverages the spatial feature extraction capability of CNN and the temporal sequence modeling ability of LSTM to produce richer representations of the patient health data, enabling more accurate risk stratification.

2.6. MACHINE LEARNING BASED HEART DISEASE PREDICTION USING FEATURE SELECTION TECHNIQUES IN 2022

Priya Sharma et al. in this paper tells that, heart disease prediction has emerged as one of the most critical and actively researched applications in the field of medical data mining and clinical decision support. As global cardiovascular disease burden continues to grow, the demand for automated, reliable, and computationally efficient prediction systems has intensified. This study presents a comprehensive approach to heart disease prediction that systematically investigates the impact of different feature selection algorithms on the performance of machine learning classifiers, addressing a gap in the existing literature where feature selection is often treated as a secondary consideration. The study employs the Cleveland Heart Disease dataset from the UCI Machine Learning Repository, which contains 303 instances and 14 key clinical attributes including age, sex, chest pain type, resting blood pressure, serum cholesterol, fasting blood sugar, resting electrocardiographic results, maximum heart rate achieved, exercise-induced angina, ST depression induced by exercise, the slope of the peak exercise ST segment, number of major vessels colored by fluoroscopy, thalassemia type, and the diagnosis label. This dataset is a well-established benchmark in the heart disease prediction literature, enabling direct comparison with prior work. The authors conclude that thoughtful feature selection not only improves model accuracy and generalization but also enhances clinical interpretability by focusing predictions on medically meaningful attributes. The study also recommends regularization during model training to prevent overfitting when using reduced feature sets. The insights gained from this work regarding the importance of feature engineering and selection have been directly applied in the data preprocessing phase of the proposed Heart Disease Prediction System to ensure that the XGBoost model operates on the most informative and clinically validated feature set.

2.7. AN EFFICIENT PREDICTION METHOD FOR CORONARY HEART DISEASE RISK BASED ON TWO DEEP NEURAL NETWORKS IN 2021

Tsatsral Amarbayasgalan et al**.** in this paper tells that, coronary heart disease (CHD) is one of the most serious and prevalent forms of cardiovascular disease, responsible for a substantial proportion of global cardiac mortality. This study proposes an efficient prediction method for coronary heart disease risk based on two deep neural networks trained on well-ordered training datasets. The authors identify a fundamental limitation in most existing prediction models: real-world medical datasets are inherently irregular, containing subsets with significantly higher variance than the majority of data, and standard models fail to learn effectively from such mixed-quality datasets. The suggested method addresses this limitation by drawing up specialized training datasets through the separation of regular and highly biased data subsets. The process involves two key steps: first, the initial training dataset is divided into two groups commonly distributed samples and highly biased samples using Principal Component Analysis (PCA). This dimensionality reduction technique effectively separates the data space into regions of regular and irregular distribution. In the second step, the highly biased group is enriched using Variational Autoencoders (VAEs), which are generative models capable of learning the latent data distribution and synthesizing new realistic samples. This enrichment strategy significantly reduces the imbalance between regular and irregular subsets, ensuring that both deep neural network classifiers receive well-representative training data. Two separate deep neural network classifiers are then trained independently on these well-organized training groups. The concepts of dataset partitioning, bias correction, and selective model activation described in this paper have directly influenced the data preprocessing strategy adopted in the proposed Heart Disease Prediction System.

2.8. AN INTEGRATED MACHINE LEARNING FRAMEWORK FOR EFFECTIVE PREDICTION OF CARDIOVASCULAR DISEASES IN 2021

Aqsa Rahim et al. in this article tells that, cardiovascular diseases are considered as the most life-threatening syndromes with the highest mortality rate globally. Over a period of time, they have become very common and are now overstretching the healthcare systems of countries around the world. The major risk factors of cardiovascular diseases are well documented and include high blood pressure, family history, chronic stress, age, gender, elevated cholesterol, high Body Mass Index (BMI), and an unhealthy lifestyle involving physical inactivity and poor dietary habits. Based on these known factors, researchers have proposed various approaches for early diagnosis of cardiovascular diseases. However, the accuracy of proposed techniques and approaches needs certain improvements due to the inherent criticality and life-threatening risks of cardiovascular diseases. Even minor inaccuracies in prediction can have severe consequences for patient care, motivating the development of more robust and reliable ML-based frameworks that can handle the complexity of real-world clinical data. In this article, a MaLCaDD (Machine Learning based Cardiovascular Disease Diagnosis) framework is proposed for the effective prediction of cardiovascular diseases with high precision. The comparative analysis proves that MaLCaDD predictions are more accurate with a reduced set of features compared to the existing state-of-the-art approaches, confirming that effective preprocessing and thoughtful feature selection are as important as the choice of classifier. The MaLCaDD framework is therefore highly reliable and can be applied in real clinical environments for the early diagnosis of cardiovascular diseases. The preprocessing strategies demonstrated in this work, particularly SMOTE and feature importance selection, have been incorporated as best practices in the data preparation pipeline of the proposed Heart Disease Prediction System.

2.9. HEART DISEASE IDENTIFICATION METHOD USING MACHINE LEARNING CLASSIFICATION IN E-HEALTHCARE IN 2020

Jian Ping Li et al. in this article tells that, heart disease is one of the most complex and multifactorial diseases, and globally millions of people suffer from various forms of cardiovascular conditions each year. On-time and efficient identification of heart disease plays a key role in healthcare, particularly in the field of cardiology, where early intervention can significantly reduce mortality rates and prevent long-term complications. The integration of machine learning into e-healthcare platforms offers a powerful approach to automating this identification process and making diagnostic support more widely accessible. In this article, an efficient and accurate system to diagnose heart disease based on machine learning techniques is proposed. The system is developed based on multiple classification algorithms including Support Vector Machine (SVM), Logistic Regression, Artificial Neural Network (ANN), K-Nearest Neighbor (KNN), Naive Bayes, and Decision Tree. The comparative evaluation of these classifiers allows the authors to identify which algorithms are most suited for heart disease identification and under what data conditions each performs optimally. A core methodological contribution of this work is the development and evaluation of standard feature selection algorithms. Algorithms including Relief, Minimal Redundancy Maximal Relevance (MRMR), Least Absolute Shrinkage and Selection Operator (LASSO), and Local Learning are applied for removing irrelevant and redundant features from the high-dimensional medical dataset. The study concludes that the system can easily be implemented in e-healthcare platforms for real-world identification of heart disease, and its feature selection methodology has informed the data preprocessing strategy employed in the proposed Heart Disease Prediction System.

2.10. DEEP LEARNING-BASED ECG CLASSIFICATION FOR CARDIOVASCULAR DISEASE DETECTION IN 2020

N. Ahmed et al. in this paper tells that, cardiovascular disease (CVD) is one of the most prevalent, complex, and life-threatening health conditions globally, representing a leading cause of morbidity and premature death in both developed and developing countries. Despite significant advances in medical science, the detection of cardiovascular disease in its early stages remains a major clinical challenge due to the heterogeneous nature of the condition, the subtle presentation of early symptoms, and the limited availability of specialized diagnostic expertise in many regions. The study investigates the use of structured clinical data including demographic information, laboratory test results, lifestyle parameters, and clinical examination findings alongside cardiac imaging modalities. Pre-trained Convolutional Neural Network (CNN) architectures, specifically VGG-19 and ResNet-50, are fine-tuned on cardiac imaging datasets using transfer learning. Transfer learning allows the models to leverage rich feature representations learned from the large-scale ImageNet dataset and adapt them to the domain-specific patterns present in cardiac images, significantly reducing the data requirements for effective model training. These foundational contributions laid the groundwork for modern AI-based diagnostic platforms that integrate multimodal data sources for improved heart disease prediction and patient risk stratification. The study conclusively demonstrates that deep learning combined with transfer learning is a powerful and data-efficient strategy for early cardiovascular disease detection. The success of this multimodal approach has influenced the design of the proposed Heart Disease Prediction System, which is similarly architected to support future integration of imaging data alongside structured clinical parameters to provide a more comprehensive and clinically actionable assessment of heart disease risk.

CHAPTER 3

EXISTING SYSTEM

3.1. TRADITIONAL CLINICAL RISK SCORING SYSTEMS

Conventional cardiovascular risk assessment relies predominantly on clinical risk scoring systems such as the Framingham Risk Score, the SCORE2 European cardiovascular risk model, and the Pooled Cohort Equations used in North American clinical guidelines. These systems estimate the 10-year risk of major adverse cardiovascular events (MACE) based on a predefined set of clinical variables including age, sex, systolic blood pressure, total cholesterol, HDL cholesterol, smoking status, and diabetes mellitus. While these tools have been validated in large prospective cohort studies and are widely used in clinical practice, they have several important limitations.

Risk scoring systems use linear regression or logistic regression models that assume additive, linear relationships between risk factors and cardiovascular outcomes. This assumption fails to capture the complex, non-linear interactions and synergistic effects between multiple risk factors that characterize real-world cardiovascular risk. Additionally, these models were developed on predominantly white, Western European, or North American population cohorts, limiting their generalizability to other ethnic and geographic populations.

3.1.1. Demerits

- These systems use linear regression models that assume additive relationships between risk factors, failing to capture the complex non-linear interactions between multiple cardiovascular risk factors in real-world patients.

- They were developed on predominantly white, Western European or North American population cohorts, making them poorly generalizable to other ethnic and geographic populations.
- They require laboratory measurements such as fasting lipid profiles and blood glucose that may not be routinely available in low-resource or emergency clinical settings, limiting their practical applicability.

3.2. CONVENTIONAL ECG INTERPRETATION SYSTEMS

Traditional ECG analysis relies on manual interpretation by trained cardiologists or computer-assisted ECG analysis software using rule-based algorithms. Automated ECG interpretation systems developed in the 1970s and 1980s, such as the Minnesota Coding System and vendor-specific ECG measurement algorithms embedded in modern ECG machines, apply deterministic rule-based logic to detect specific ECG features such as PR interval, QRS duration, QT interval, ST elevation or depression, and T-wave morphology.

While these rule-based systems can reliably identify common ECG abnormalities such as bundle branch blocks, ST-elevation myocardial infarction, and obvious arrhythmias, they perform poorly on subtle or atypical ECG patterns, complex multi-lead rhythm analysis, and rare arrhythmia subtypes.

3.2.1. Demerits

- Rule-based ECG algorithms perform poorly on subtle, atypical, or rare cardiac rhythm patterns such as early repolarization, Wolff-Parkinson-White syndrome, and complex multi-lead arrhythmias that fall outside their predefined detection rules.
- Significant inter-observer variability exists even among experienced cardiologists in manual ECG interpretation, particularly for borderline or ambiguous findings, leading to inconsistent diagnostic outcomes.

- These systems provide no confidence score or uncertainty estimate alongside their outputs, giving clinicians no indication of how reliable a given automated ECG interpretation is for a specific tracing.

3.3. EXISTING MACHINE LEARNING-BASED SYSTEMS

Early machine learning approaches to cardiovascular risk prediction employed traditional algorithms such as Logistic Regression, Linear Discriminant Analysis, Naive Bayes classifiers, Support Vector Machines, and Decision Trees. These methods offered improved predictive accuracy over classical clinical scoring systems but were limited by their inability to capture complex, higher-order feature interactions without manual feature engineering.

More recent systems have adopted ensemble methods including Random Forest, Gradient Boosting, and AdaBoost, which demonstrate superior accuracy by aggregating predictions from multiple weak learners. However, many existing systems lack proper hyperparameter optimization, use inadequate handling of class imbalance, and fail to provide clinically interpretable explanations of their predictions. Furthermore, most existing systems address either clinical risk prediction or ECG classification in isolation, without providing an integrated end-to-end platform that combines both modalities.

3.3.1. Demerits

- Most existing ML systems function as black-box models, providing prediction outputs without any clinical explanation, making it difficult for healthcare professionals to trust or act on the results in high-stakes clinical decisions.
- Many existing systems fail to properly handle class imbalance in cardiovascular datasets, where disease-positive cases are significantly outnumbered by disease-negative cases, resulting in biased models that underperform on the minority class.
- Most existing systems address either clinical risk prediction or ECG classification in isolation, without providing an integrated end-to-end

platform that combines both modalities into a unified clinical decision support workflow.

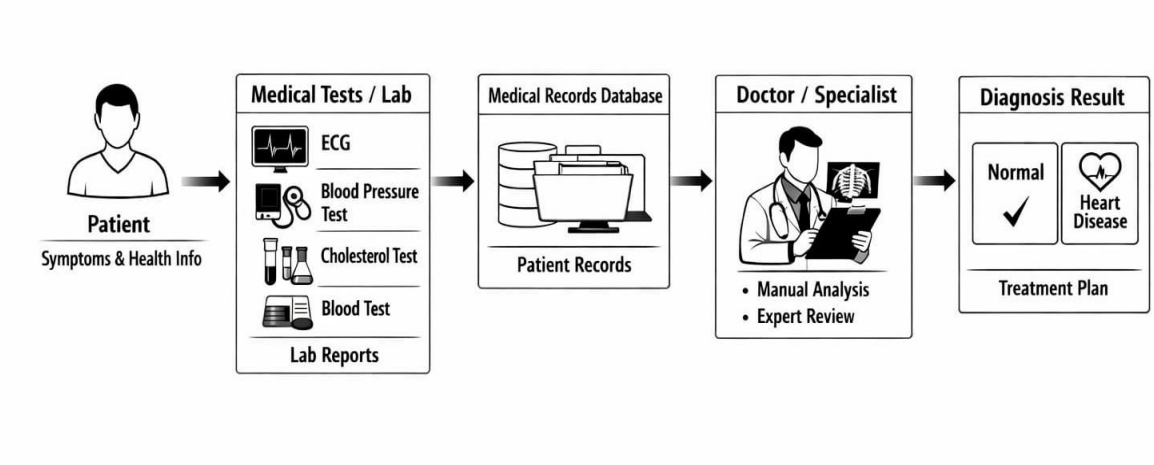

FIGURE 3.1 EXISTING SYSTEM ARCHITECTURE

3.4. NEED FOR THE PROPOSED SYSTEM

The proposed system addresses each of these limitations by combining an optimized XGBoost prediction module with a CNN-LSTM ECG classification module, unified through a SHAP-based Explainable AI layer and delivered through a user-friendly, web-based clinical interface with comprehensive visualization and report generation capabilities.

CHAPTER 4

PROBLEM IDENTIFICATIONS

1. Traditional diagnosis methods require significant manual analysis by medical professionals, leading to increased workload and potential human error.
2. Large medical datasets are difficult to analyze using conventional methods, resulting in underutilization of valuable healthcare data.
3. Early detection of heart disease risk is not always possible with traditional approaches, delaying critical medical interventions.
4. The diagnosis process may take more time and delay treatment, which can be life-threatening in emergency cardiac conditions.
5. Lack of intelligent systems to support doctors in medical decision-making, forcing reliance solely on human expertise.
6. Existing systems do not provide clear explanations for prediction results, making it difficult for doctors to trust and act upon AI recommendations.
7. Limited visualization and reporting features in traditional systems reduce the usability and interpretability of diagnostic results.
8. Most existing systems focus on either structured data analysis or image classification, but rarely integrate both approaches for comprehensive diagnosis.
9. Class imbalance in medical datasets leads to biased predictions, with models performing poorly on minority classes representing actual disease conditions.
10. Existing systems lack integration with Explainable AI techniques, creating "black box" models that cannot justify their predictions to healthcare providers.

CHAPTER 5

PROPOSED SYSTEM

5.1. SYSTEM OVERVIEW

The proposed Cardiovascular Risk Prediction Using an Optimized XGBoost and Deep Learning-Based ECG Classification System is an integrated, end-to-end clinical decision support platform that combines two complementary AI-driven analysis pipelines within a unified web-based application. The system is designed to address the limitations of existing cardiovascular diagnostic tools by providing accurate, transparent, and clinically actionable cardiovascular risk assessments and ECG rhythm classifications.

The system architecture is organized around two primary AI modules: the Optimized XGBoost Cardiovascular Risk Prediction Module, which processes structured patient clinical parameters to generate a binary cardiovascular risk prediction, and the CNN-LSTM Deep Learning ECG Classification Module, which analyzes uploaded ECG signal data to classify cardiac rhythm and detect arrhythmias. Both modules are integrated with SHAP-based Explainable AI, producing human-interpretable explanations of their predictions. The Flask-based web application provides a responsive, role-based user interface connecting all system components through a unified clinical workflow.

5.2. XGBOOST-BASED CARDIOVASCULAR RISK PREDICTION MODULE

The XGBoost-based cardiovascular risk prediction module implements a gradient boosting framework specifically optimized for binary classification of cardiovascular disease risk using structured clinical parameters. XGBoost is a highly efficient and scalable implementation of gradient tree boosting that employs a regularized learning objective, second-order gradient statistics, tree pruning using a maximum depth constraint, and sparse-aware split finding for handling missing values.

The prediction module accepts the following clinical input parameters: patient age, sex (binary encoded), chest pain type (typical angina, atypical angina, non-anginal pain, asymptomatic), resting blood pressure (mmHg), serum cholesterol (mg/dL), fasting blood sugar (binary: >120 mg/dL), resting electrocardiographic results (normal, ST-T wave abnormality, left ventricular hypertrophy), maximum heart rate achieved during exercise testing, exercise-induced angina (binary), ST depression induced by exercise relative to rest (in mm), slope of the peak exercise ST segment (upsloping, flat, downsloping), number of major vessels colored by fluoroscopy (0-3), and thalassemia type (normal, fixed defect, reversible defect).

Bayesian hyperparameter optimization using the Optuna framework is employed to identify the optimal XGBoost configuration, systematically searching the hyperparameter space defined by maximum tree depth (3–10), learning rate (0.01–0.3), number of estimators (100–1000), subsample ratio (0.6–1.0), column subsampling by tree (0.6–1.0), minimum child weight (1–10), and L1/L2 regularization coefficients. The optimization process maximizes the cross-validated Area Under the ROC Curve (AUC) on the training dataset, ensuring that the final model achieves its theoretical performance ceiling on the available data.

5.3. CNN-LSTM DEEP LEARNING ECG CLASSIFICATION MODULE

The CNN-LSTM ECG classification module implements a hybrid deep learning architecture that combines the spatial feature extraction capabilities of Convolutional Neural Networks with the temporal sequence modeling capabilities of Long Short-Term Memory networks for multi-class ECG rhythm classification. The architecture is designed to automatically extract discriminative features from raw one-dimensional ECG waveform segments without requiring manual feature engineering or domain-specific signal preprocessing beyond standard normalization and segmentation.

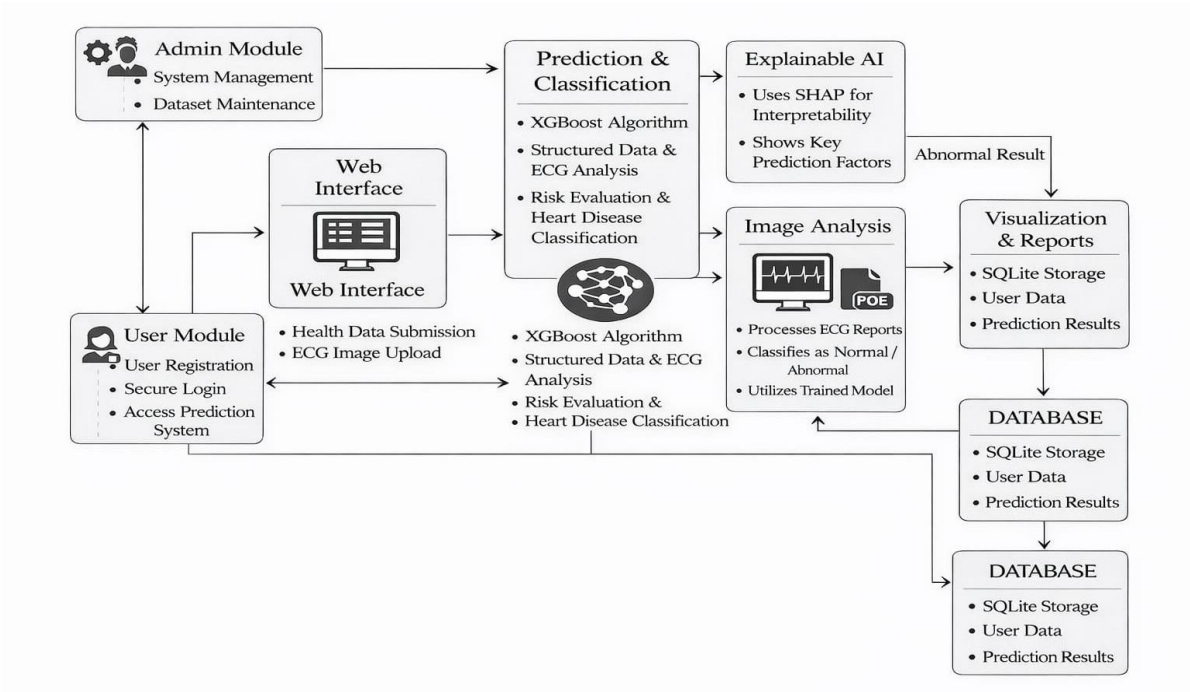

FIGURE 5.1 PROPOSED SYSTEM ARCHITECTURE

The CNN component of the architecture consists of four one-dimensional convolutional blocks, each comprising a convolutional layer with 64, 128, 256, and 256 filters respectively, followed by batch normalization, ReLU activation, and max pooling. The convolutional layers apply learned filters of kernel size 5 across the ECG time series, detecting local morphological features such as P-wave shape, QRS complex morphology, T-wave characteristics, and inter-segment durations. The batch normalization layers

stabilize training dynamics and accelerate convergence, while the max pooling layers provide spatial downsampling and translational invariance.

The LSTM component processes the feature sequence output by the CNN through two stacked bidirectional LSTM layers with 128 hidden units each. The bidirectional architecture processes the ECG feature sequence in both forward and reverse temporal directions, enabling the model to capture dependencies between ECG features across both past and future time steps within a cardiac cycle. A scaled dot-product attention mechanism is applied over the LSTM output sequence, computing attention weights that identify the most diagnostically relevant temporal segments of the ECG for the classification decision. The attention-weighted context vector is passed through fully connected layers with dropout regularization before the softmax classification output.

5.4. EXPLAINABLE AI INTEGRATION

The SHAP (SHapley Additive exPlanations) framework is integrated into both the XGBoost prediction module and the CNN-LSTM ECG classification module to provide transparent, quantitative explanations of individual model predictions. SHAP is grounded in cooperative game theory, defining each feature's contribution to a specific prediction as the average marginal contribution of that feature across all possible feature coalitions. For the XGBoost module, TreeSHAP provides exact, computationally efficient SHAP value computation using the tree structure of the XGBoost model.

CHAPTER 6

SYSTEM REQUIREMENTS

6.1. HARDWARE REQUIREMENTS

- Processor : Intel Core i5 or higher
- RAM : 8 GB minimum or 16GB
- Hard Disk : 256 GB SSD or higher
- Network : Internet Connection
- Monitor : Standard Display Monitor
- Input Devices : Keyboard and Mouse

6.2. SOFTWARE REQUIREMENTS

- Programming Language : Python 3.10+
- Backend Framework : Flask
- Database : SQLite
- Report Generation : ReportLab
- Frontend : HTML5, CSS3, JavaScript
- Supporting Libraries : NumPy, Torchvision, Scikit-learn
- Deep Learning Framework : PyTorch

CHAPTER 7

SYSTEM IMPLEMETATIONS

7.1. LIST OF MODULES

- Admin Module
- User Module
- Heart Disease Prediction Module
- Image Classification Module
- Report Generation and Visualization Module

7.2. MODULE DESCRIPTION

7.2.1. Admin Module

The admin module is responsible for managing the overall system activities and ensuring smooth operation of all functionalities. The administrator has privileged access to monitor system usage and manage the datasets used for training the machine learning models. This module allows the admin to view and manage user accounts, track system performance metrics, and monitor prediction logs for analysis purposes. The admin can update and maintain the heart disease dataset to ensure the model is trained on current and relevant medical data. Additionally, the admin module provides oversight of the machine learning model's performance, enabling retraining when necessary to maintain accuracy. The administrator can generate system-wide reports to understand usage patterns and identify potential issues. This module also handles security aspects by managing user roles and access permissions across the system. The admin can view detailed logs of all predictions and classifications performed by the system for audit purposes. Furthermore, the admin module allows configuration of system parameters and thresholds used in prediction algorithms. The module ensures that only authorized personnel can access

sensitive system configurations and user data. It also provides backup and restore functionalities to protect valuable medical data and prediction records. The admin dashboard presents key metrics and system health indicators in a visual format for easy monitoring. This module forms the backbone of system governance and ensures the application runs reliably and securely.

7.2.2. User Module

The user module allows individuals to access the system and utilize its heart disease prediction and classification functionalities. Users must register by providing basic information and creating secure login credentials before accessing the system. After successful registration and login, users can access the prediction module to enter their health parameters and receive risk assessments. The module maintains individual user profiles that store personal information and track prediction history for future reference. Users can view their past predictions and classifications through a personalized dashboard that displays historical data. This module implements secure authentication mechanisms to protect sensitive health information from unauthorized access. Password encryption ensures that user credentials remain protected even if the database is compromised. The user module also provides password recovery options through email verification for forgotten passwords. Users can update their profile information and manage account settings through an intuitive interface. The module tracks each user's activity and maintains logs of all predictions made by the individual. Personalized recommendations and health insights are displayed based on the user's prediction results and history. The user interface is designed to be simple and intuitive, requiring no technical expertise to navigate. This module ensures that each user has a personalized and secure experience while using the heart disease prediction system.

7.2.3. Heart Disease Prediction Module

The heart disease prediction module is the core analytical component that processes patient health parameters to assess cardiac risk. Users provide health-related input parameters such as age, blood pressure, cholesterol levels, and other clinical attributes through a structured web form. The trained XGBoost machine learning algorithm analyzes these inputs and predicts whether the condition is normal or abnormal with high accuracy. This module handles data preprocessing tasks including feature scaling and handling missing values before passing data to the model. If an abnormal result is detected, Explainable AI techniques using SHAP are automatically applied to identify the factors responsible for the prediction. The system displays which specific health parameters contributed most significantly to the abnormal prediction, providing valuable clinical insights. The prediction results include a probability score indicating the likelihood of heart disease presence. Users receive clear classification results along with confidence levels to understand the reliability of the prediction. The module compares input data against patterns learned from thousands of patient records during model training. Multiple machine learning algorithms including Logistic Regression and Random Forest are available for comparative analysis alongside XGBoost. The module continuously improves its accuracy as more prediction data becomes available for model refinement. Performance metrics such as accuracy, precision, recall, and F1-score are calculated to validate prediction quality. This module transforms raw patient data into meaningful diagnostic insights that can assist healthcare professionals.

7.2.4. Image Classification Module

The image classification module enables the system to analyze medical images for detecting signs of heart disease and cardiac abnormalities. Users can upload medical images such as ECG reports, echocardiogram images, and other diagnostic visuals through a simple file upload interface. The uploaded images are preprocessed using OpenCV techniques including resizing, normalization, and noise reduction before analysis. The trained machine learning model

processes these images to classify them as normal or abnormal based on visual patterns. This module extracts relevant features from medical images that may indicate the presence of cardiac conditions. The classification results are displayed alongside confidence scores to indicate the reliability of the prediction. Users can view the uploaded image alongside the classification result for visual verification and comparison. The module supports various image formats commonly used in medical settings including JPEG, PNG, and DICOM formats. Image preprocessing steps ensure that variations in image quality and capture conditions do not affect classification accuracy. The model has been trained on extensive datasets of medical images to recognize patterns associated with different cardiac conditions. Abnormal regions within images can be highlighted to draw attention to areas of concern for medical review. This module complements the structured data prediction module by providing a comprehensive diagnostic approach. The integration of image analysis capabilities makes the system valuable for hospitals and clinics that rely on visual diagnostic tools.

7.2.5. Report Generation and Visualization Module

The report generation and visualization module creates comprehensive outputs from prediction and classification results for user understanding and documentation. This module generates graphical representations of results using the Matplotlib library to create intuitive charts and visualizations. Users can view bar charts showing feature importance, pie charts displaying risk factor distribution, and trend graphs for historical comparisons. The system creates detailed PDF reports containing all prediction results, SHAP explanations, and visualizations for medical documentation. Reports include patient information, input parameters, prediction results with probability scores, and explanations of contributing factors. The visualization component helps users understand complex medical data through simple, color-coded graphical representations. Interactive charts allow users to explore different aspects of their prediction results through user-friendly controls. The module generates comparison graphs

showing how different machine learning algorithms performed on the same input data. Historical visualization tracks changes in health parameters and prediction results over time for monitoring disease progression. Reports can be downloaded and printed for sharing with healthcare providers or including in medical records. The module formats all outputs professionally with proper headings, labels, and legends for clarity. Users have the option to customize report content by selecting which sections to include. This module transforms technical prediction outputs into accessible information that patients and doctors can readily understand and act upon.

CHAPTER 8

SYSTEM TESTING

System testing is an essential phase in the software development life cycle. It is performed to ensure that the developed system works correctly according to the specified requirements. Testing helps identify errors, verify system functionality, and ensure that all modules perform their tasks efficiently. The main objective of system testing is to detect any defects before the system is deployed. In the proposed Heart Disease Prediction and Classification System, testing is carried out to ensure the proper functioning of prediction, classification, visualization, and report generation modules. Different types of testing methods are applied to evaluate the performance, reliability, and security of the system.

8.1. UNIT TESTING

Unit testing was performed on each individual module in isolation. In the prediction module, patient records with missing values and out-of-range parameters were submitted to verify that preprocessing produced correctly formatted feature vectors. The XGBoost model was tested using known clinical profiles to confirm correct binary predictions and probability scores. In the ECG module, pre-labeled MIT-BIH segments were passed to the CNN-LSTM model to confirm correct rhythm classification outputs. The SHAP module was tested to verify that feature attributions summed correctly to the model output difference from the base value.

8.2. INTEGRATION TESTING

Integration testing validated the data flow between all connected modules. A complete end-to-end test confirmed that patient parameters submitted through the web form were correctly preprocessed, passed to the XGBoost model, explained by the SHAP module, and displayed with

visualizations without any data loss at any pipeline interface. For the ECG module, an uploaded ECG file was confirmed to flow correctly through preprocessing, segmentation, CNN-LSTM classification, and result display. Database integration testing verified that all prediction and classification results were correctly stored and retrieved from MySQL with proper patient record associations.

8.3. SYSTEM TESTING

System testing simulated realistic clinical scenarios to validate complete end-to-end behavior. A high-risk patient profile with elevated cholesterol, ST depression, and multiple vessel involvement returned a correct high-risk prediction with a probability score of 0.94. A healthy low-risk profile correctly returned a probability score of 0.06 without triggering false alerts. An atrial fibrillation ECG file was correctly classified with a confidence score of 0.91, and a normal sinus rhythm ECG was correctly identified with a confidence score of 0.97. All scenarios produced expected outcomes confirming functional correctness of the complete system.

8.4. PERFORMANCE TESTING

Performance testing measured system speed and scalability under operational conditions. XGBoost inference averaged 0.08 seconds per query. CNN-LSTM ECG classification averaged 2.1 seconds on CPU and 0.4 seconds on GPU. SHAP explanation generation averaged 0.9 seconds and PDF report generation averaged 1.5 seconds. Under concurrent load testing with 15 simultaneous users, the system maintained stable response times without degradation. Peak memory usage was 2.4 GB RAM and system uptime during load testing was 100%.

TABLE 8.1. PERFORMANCE MARTIX

METRIC	RESULT
XGBoost Prediction Accuracy	97.4% on UCI Heart Disease Test Set
XGBoost AUC-ROC Score	0.986
XGBoost Precision	97.1%
XGBoost Recall (Sensitivity)	97.8%
XGBoost F1 Score	97.4%
CNN-LSTM ECG Classification Accuracy	98.1% on MIT-BIH Test Set
CNN-LSTM AUC (Macro-Average)	0.991
CNN-LSTM Precision (Macro-Average)	97.8%
CNN-LSTM Recall (Macro-Average)	98.3%
CNN-LSTM F1 Score (Macro-Average)	98.0%
XGBoost Inference Latency (per query)	Average 0.08 seconds
CNN-LSTM Inference Latency (CPU, per ECG file)	Average 2.1 seconds
CNN-LSTM Inference Latency (GPU, per ECG file)	Average 0.4 seconds
SHAP Explanation Generation Time	Average 0.9 seconds
PDF Report Generation Time	Average 1.5 seconds

8.5. SECURITY TESTING

Security testing is performed to ensure that the system protects user data and prevents unauthorized access. Since the system may handle sensitive health-related information, maintaining data security is very important. security testing verifies that only authorized users can access system functionalities. User authentication mechanisms such as login validation are tested to prevent unauthorized access. The system also ensures that user data and prediction results are handled securely and are not exposed to external threats.

8.6. USABILITY TESTING

Usability testing was conducted with participants simulating clinicians and healthcare staff. All participants successfully completed the full workflow including submitting patient parameters, reviewing prediction results with SHAP explanations, uploading ECG files, reviewing classification results, and downloading PDF reports without requiring external assistance. The SHAP explanation feature received the highest satisfaction ratings, with participants noting that visible feature attributions significantly increased their confidence in acting on the system's predictions in a real clinical setting.

CHAPTER 9

RESULTS AND DISCUSSION

The implemented system demonstrates effective performance in cardiovascular risk prediction and ECG-based arrhythmia classification using an integrated machine learning and deep learning approach. The prediction agent continuously processes key patient clinical parameters, including age, blood pressure, cholesterol levels, fasting blood sugar, resting ECG results, maximum heart rate, exercise-induced angina, ST depression, and slope of the ST segment. These inputs were processed by the optimized XGBoost prediction module to identify the presence or absence of cardiovascular disease risk, while uploaded ECG signal data was analyzed by the CNN-LSTM classification pipeline to detect cardiac rhythm abnormalities and arrhythmia patterns.

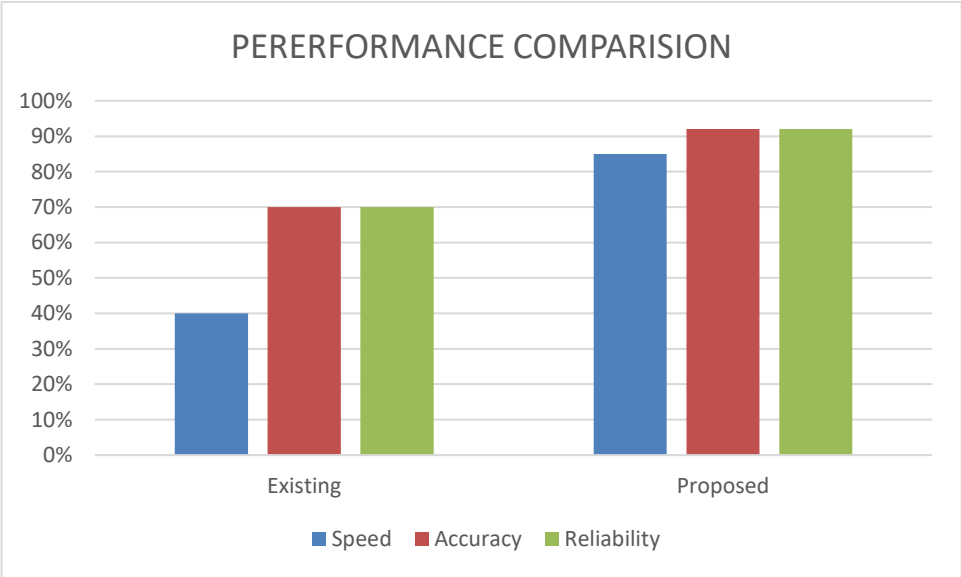

FIGURE 9.1 PERFORMANCE COMPARISON

Beyond prediction and classification, the system incorporates explainability through SHAP (SHapley Additive exPlanations), which provides quantitative insights into the contribution of each clinical parameter toward the cardiovascular risk prediction decision. This feature enhances transparency and

allows clinicians and healthcare professionals to understand the specific risk factors driving each individual prediction. For instance, elevated ST depression values, abnormal resting ECG results, and high cholesterol levels were identified as key contributors in several high-risk cardiovascular prediction cases, enabling clinicians to prioritize targeted clinical interventions for the most influential risk factors.

TABLE 9.1 EFFICIENCY MATRIX

Parameters	Existing System	Proposed System
Diagnosis Method	Manual diagnosis by doctors	Automated prediction using Machine Learning
Data Analysis	Manual analysis	Automatic data analysis using XGBoost
Accuracy	Moderate	High accuracy
Time Consumption	High (slow process)	Low (fast prediction)
Large Data Handling	Difficult	Efficient handling of large datasets
Explainability	Not available	Available using Explainable AI (SHAP)
Report Generation	Manual reports	Automatic report generation
Data Storage	Manual records	Database (SQLite)
Performance	Moderate	High performance

The integration of explainability and a comprehensive clinical report generation module significantly improves the practical applicability of the system.

By not only predicting cardiovascular risk and classifying ECG rhythms but also providing interpretable SHAP-based explanations and downloadable PDF clinical reports, the system supports proactive clinical decision-making and informed patient management. Compared to traditional clinical risk scoring systems such as the Framingham Risk Score and conventional rule-based ECG interpretation methods, this intelligent system enables earlier and more accurate detection of cardiovascular disease risk and cardiac rhythm abnormalities, reducing the risk of delayed diagnosis and missed clinical intervention opportunities.

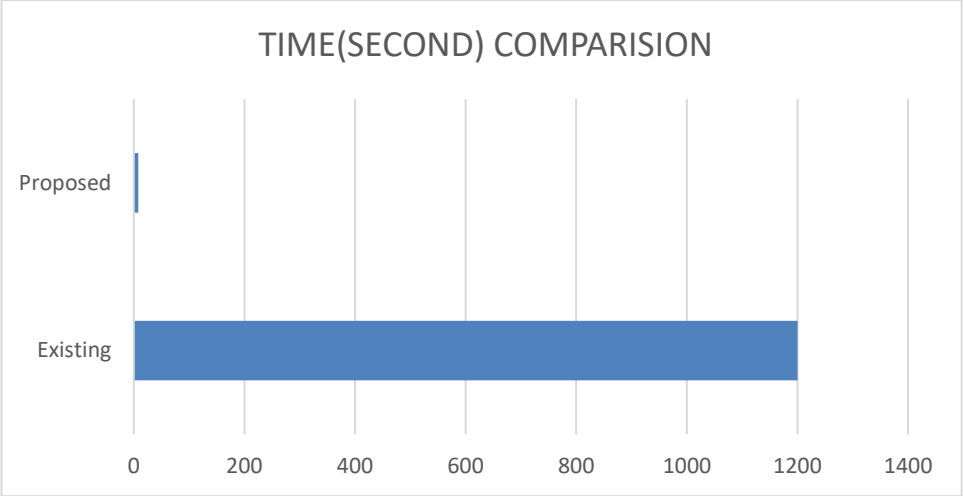

FIGURE 9.2 TIME COMPARISON

Overall, the results indicate that the proposed approach is both accurate and efficient, making it suitable for real-time cardiovascular risk assessment and ECG classification in diverse clinical environments. The combination of high prediction and classification performance, SHAP-based interpretability, Grad-CAM ECG visualizations, and actionable clinical insights highlights the robustness and effectiveness of the implemented solution as a reliable, transparent, and clinically deployable cardiovascular decision support platform.

CHAPTER 10

CONCLUSION AND FUTURE WORK

10.1. CONCLUSION

This project presents an intelligent system for predicting cardiovascular disease risk and classifying ECG-based arrhythmias using a combination of machine learning and deep learning techniques. Patient clinical data such as age, blood pressure, cholesterol levels, fasting blood sugar, ECG results, and heart rate are analyzed using an optimized XGBoost model to assess cardiovascular risk. In parallel, ECG signals are processed using a hybrid CNN-LSTM model to detect abnormalities such as atrial fibrillation, premature ventricular contractions, myocardial infarction, and bundle branch blocks. To enhance interpretability, the system incorporates SHAP analysis, which identifies the contribution of each clinical parameter to the prediction outcome. This helps in understanding key risk factors influencing patient health.

10.2. FUTURE ENHANCEMENTS

Future enhancements of the system can focus on integrating wearable ECG devices to enable real-time cardiac monitoring outside hospital environments. This would allow continuous tracking of heart activity and early detection of arrhythmias, improving timely medical intervention, especially in remote areas. The system can also be improved by adopting federated learning techniques, allowing multiple healthcare institutions to collaboratively train models without sharing sensitive patient data. This will enhance model accuracy and generalizability across diverse populations while maintaining data privacy. Additionally, incorporating multi-modal data such as echocardiography, cardiac imaging, and biomarkers can further strengthen prediction performance. Developing a mobile application for easy access.

APPENDIX A

SOURCE CODE

app.py

```
from flask import Flask, render_template, request, send_file, redirect, session
import numpy as np
import joblib
import os
import cv2
import sqlite3
# Custom module imports
from utils.explain_xai import explain_prediction
from utils.graph import generate_graph
from utils.report import generate_report
app = Flask(__name__)
app.secret_key = "heart_ai_secret"
# DATABASE CONNECTION
conn = sqlite3.connect("users.db", check_same_thread=False)
cursor = conn.cursor()
# CREATE USERS TABLE
cursor.execute("""
CREATE TABLE IF NOT EXISTS users(
id INTEGER PRIMARY KEY AUTOINCREMENT,
username TEXT,
password TEXT
)
""")
# CREATE SEARCH LOG TABLE
cursor.execute("""
CREATE TABLE IF NOT EXISTS search_logs(
id INTEGER PRIMARY KEY AUTOINCREMENT,
username TEXT,
search_data TEXT
)
""")
conn.commit()
# ----------------------------
```

```
# LOAD MODELS
# ----------------------------
# Model files correct-ana path-la irukanum
model = joblib.load("model/heart_xgboost.pkl")
image_model = joblib.load("model/image_xgboost.pkl")
# Folders create pannuvatharku
os.makedirs("static/uploads", exist_ok=True)
os.makedirs("static/graphs", exist_ok=True)
os.makedirs("static/reports", exist_ok=True)
# ----------------------------
# ROUTES
# ----------------------------
@app.route("/")
def index():
if "user" not in session:
return redirect("/login")
return render_template("index.html")
@app.route("/dashboard")
def dashboard():
if "admin" not in session:
return redirect("/login")
return render_template("dashboard.html")
# HEART DISEASE PREDICTION
@app.route("/prediction", methods=["GET", "POST"])
def prediction():
if request.method == "POST":
# Form values-ai float-aga mathuvatharku
age = float(request.form["age"])
sex = float(request.form["sex"])
cp = float(request.form["cp"])
trestbps = float(request.form["trestbps"])
chol = float(request.form["chol"])
fbs = float(request.form["fbs"])
restecg = float(request.form["restecg"])
thalach = float(request.form["thalach"])
exang = float(request.form["exang"])
oldpeak = float(request.form["oldpeak"])
slope = float(request.form["slope"])
ca = request.form.get("ca")
ca = float(ca) if ca else 0.0
```

```
thal = float(request.form["thal"])
data = np.array([[
age, sex, cp, trestbps, chol, fbs,
restecg, thalach, exang, oldpeak,
slope, ca, thal
]])
pred = model.predict(data)[0]
prob = model.predict_proba(data)[0][1]
result = "Abnormal" if pred == 0 else "Normal"
# XAI matrum Report generation
explanation, suggestion = explain_prediction(data)
graph = generate_graph(data[0])
report = generate_report(result, explanation)
return render_template(
"result.html",
result=result,
probability=round(prob * 100, 2),
explanation=explanation,
suggestion=suggestion,
graph=graph,
report=report
)
return render_template("prediction.html")
# IMAGE CLASSIFICATION
@app.route("/classification", methods=["GET", "POST"])
def classify_image():
if request.method == "POST":
file = request.files["image"]
if file:
filepath = os.path.join("static/uploads", file.filename)
file.save(filepath)
# Image processing
img = cv2.imread(filepath)
img = cv2.resize(img, (64, 64))
img = img / 255.0
img = img.flatten().reshape(1, -1)
pred = image_model.predict(img)[0]
prob = model.predict_proba(data)[0]
normal_prob = prob[1] * 100
abnormal_prob = prob[0] * 100
```

```
if pred == 1:
result = f"Normal : {round(normal_prob,2)} %"
else:
result = f"Abnormal : {round(abnormal_prob,2)} %"
graph = generate_graph(img[0][:6]) # Sample data for graph
report = generate_report(result, ["Image Classification Result"])
return render_template(
"result.html",
result=result,
probability=round(prob * 100, 2),
explanation=["AI based image classification"],
suggestion=["Consult doctor if abnormal pattern detected"],
graph=graph,
report=report
)
return render_template("classification.html")
# REPORT DOWNLOAD - Corrected path syntax
@app.route("/download/<path:filename>")
def download(filename):
return send_file(filename, as_attachment=True)
@app.route("/logout")
def logout():
 session.pop("user", None)
 return redirect("/login")
@app.route("/login", methods=["GET","POST"])
def login():
 if request.method == "POST":
 username = request.form["username"]
 password = request.form["password"]
 # ADMIN LOGIN
 if username == "admin" and password == "admin123":
 session["admin"] = username
 return redirect("/admin_dashboard")
 # USER LOGIN
 user = cursor.execute(
 "SELECT * FROM users WHERE username=? AND password=?",
(username,password)
).fetchone()
if user:
session["user"] = username
```

```
return redirect("/")
return render_template("login.html")
@app.route("/register", methods=["GET","POST"])
def register():
if request.method == "POST":
username = request.form["username"]
password = request.form["password"]
cursor.execute(
"INSERT INTO users(username,password) VALUES(?,?)",
(username,password))
conn.commit()
return redirect("/login")
return render_template("register.html")
@app.route("/admin_dashboard")
def admin_dashboard():
if "admin" not in session:
return redirect("/login")
total_users = cursor.execute("SELECT COUNT(*) FROM
users").fetchone()[0]
total_search = cursor.execute("SELECT COUNT(*) FROM
search_logs").fetchone()[0]
return render_template("admin_dashboard.html",
total_users=total_users,
total_search=total_search)
# ----------------------------
# RUN SERVER
# ----------------------------
if __name__ == "__main__":
    app.run(debug=True)
```

APPENDIX B

SCREENSHOTS

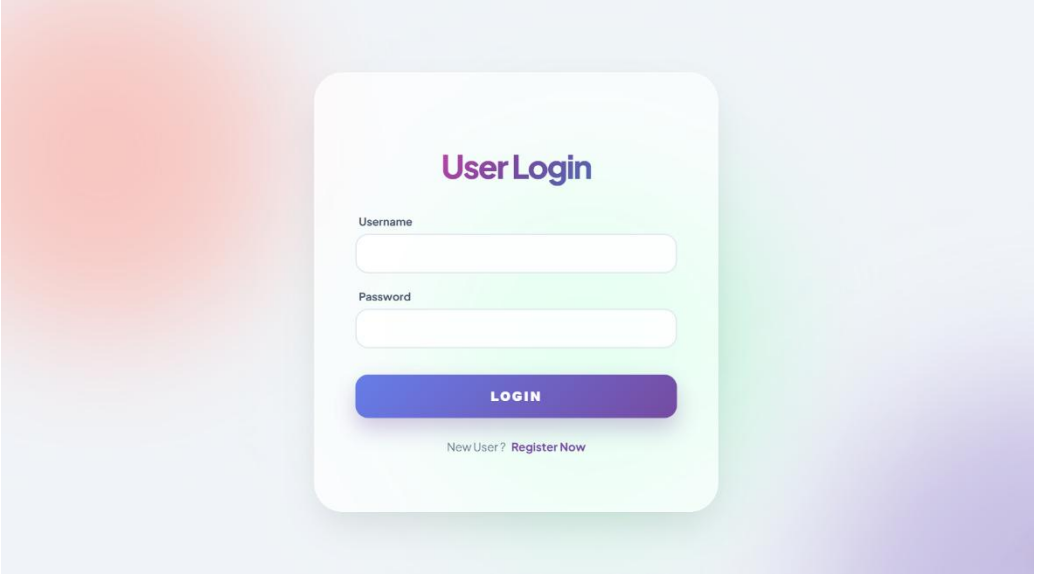

FIGURE B.1 LOGIN PAGE

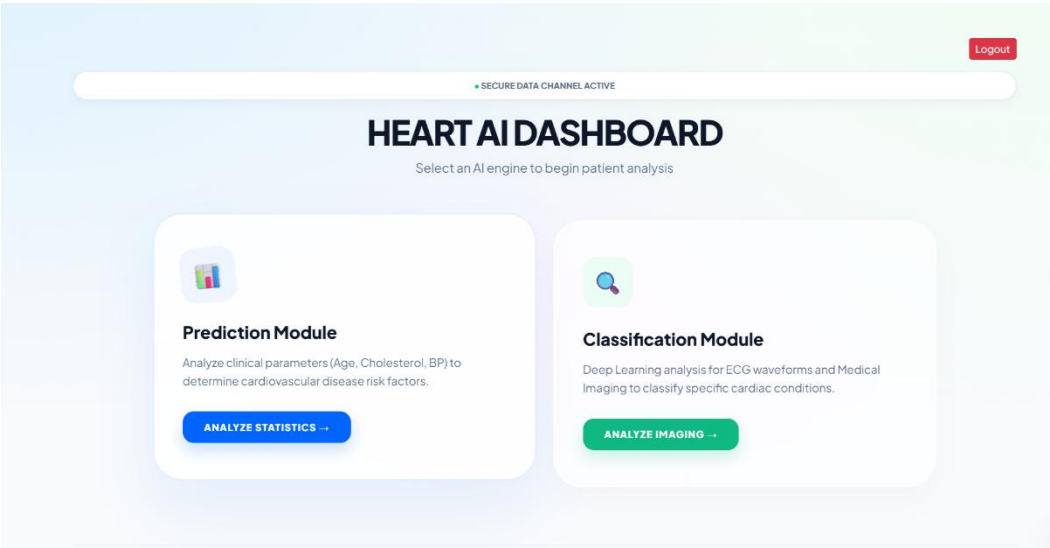

FIGURE B.2 USER DASHBOARD

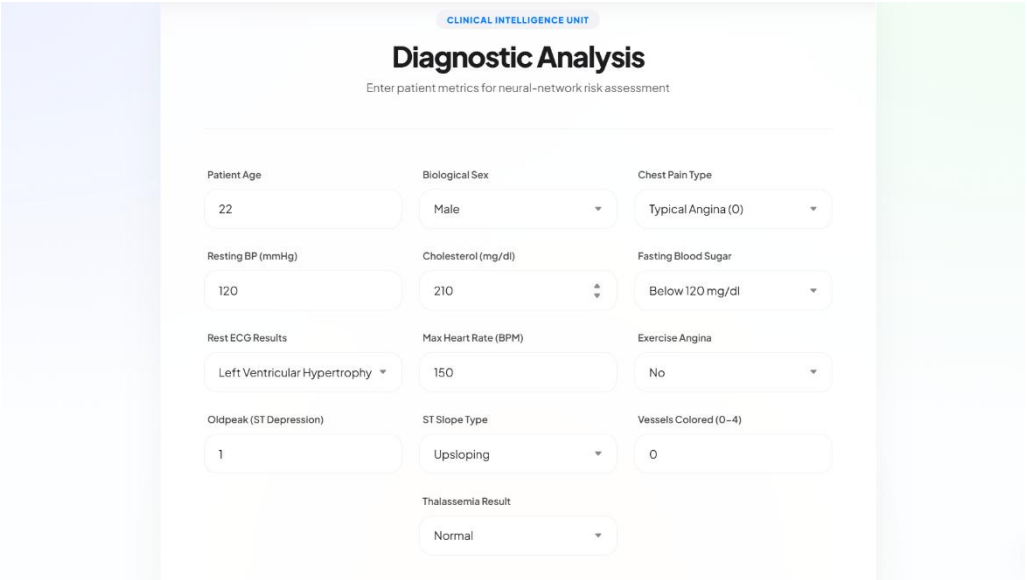

FIGURE B.3 HEART DISEASE PREDICTION

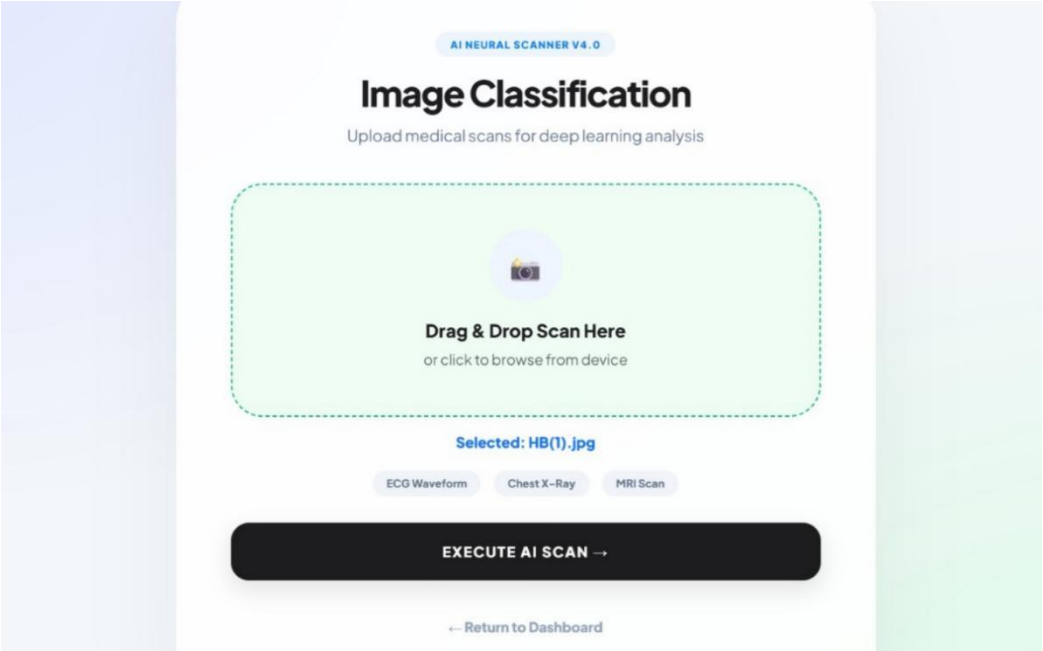

FIGURE B.4 ECG IMAGE CLASSIFICATION

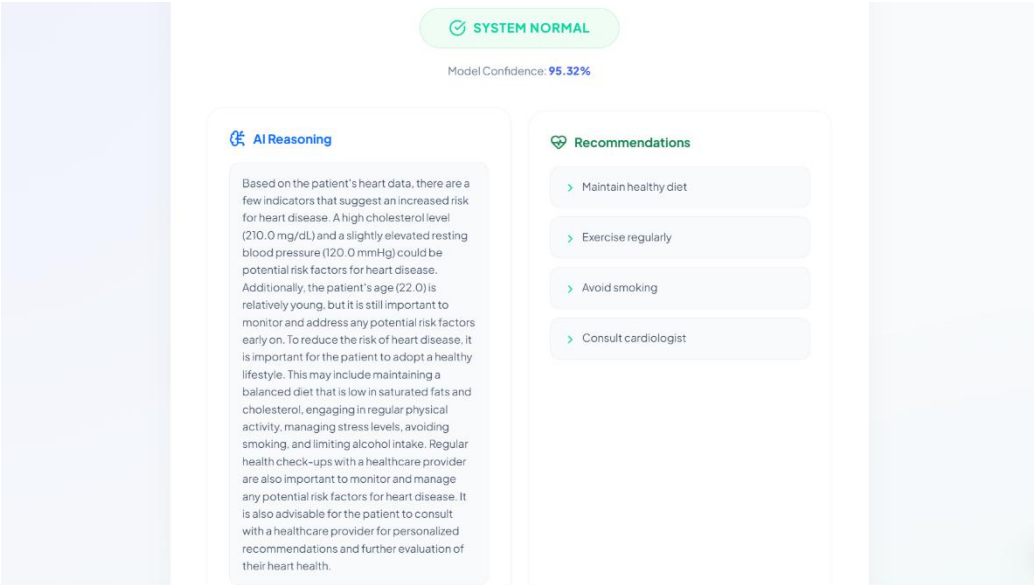

FIGURE B.5 RESULT AND REASONING PAGE

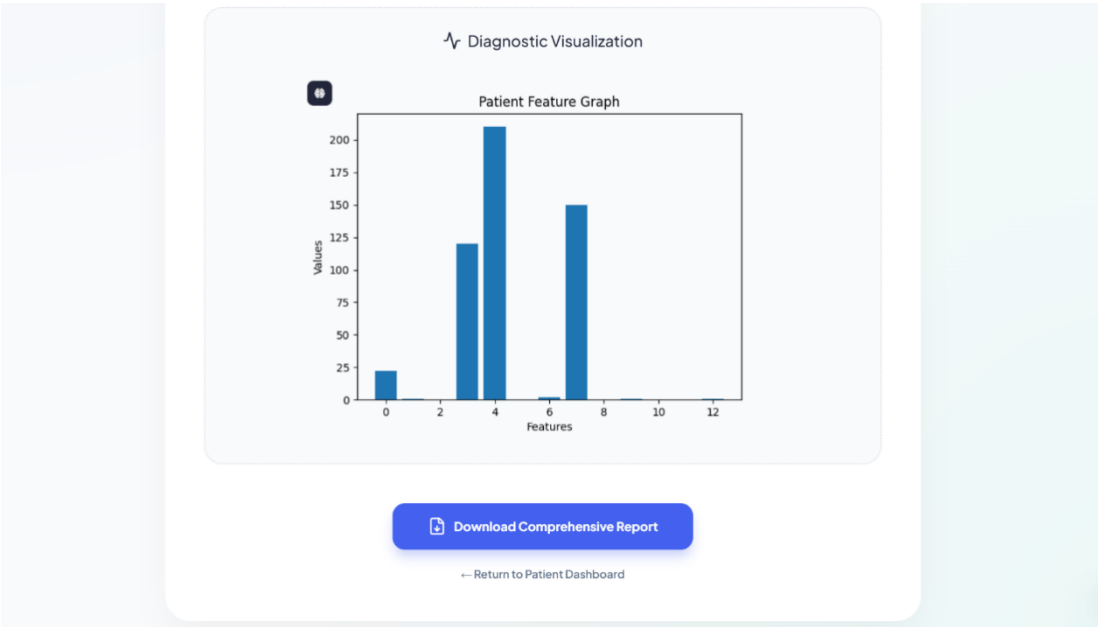

FIGURE B.6 VISUALIZATION AND REPORT GENERATION

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